

11	Which of the following statement about a geostationary satellite around Earth is true?	
	A	Its linear speed is equal to the speed of a point on the Earth's equator.
	B	It experiences zero net force as it orbits around Earth.
	C	It moves from East to West.
	D	It must remain directly above the equator.

Answer: D

Option A is wrong as the satellite's linear speed is proportional to the distance away from the centre of the earth ($\mathbf{v = r\omega}$). Hence the speed of the satellite can never be the same as the speed on the equator.

Option B is wrong as the satellite is experiencing centripetal acceleration, and thus net force as it orbits around Earth.

Option C is wrong as it moves from West to East.

Option D is correct as the geostationary satellite must be at a fixed distance directly above the Earth's equator.

12	A student heats a 500 g solid sample at an initial temperature of 10°C . The rate of heat
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